Chinese Sociological Review

ISSN: 2162-0555 (Print) 2162-0563 (Online) Journal homepage: http://www.tandfonline.com/loi/mcsa20

Social Segregation in Hong Kong's Schools: 2000–2012

Yisu Zhou, Tianji Cai & Dan Wang

To cite this article: Yisu Zhou, Tianji Cai & Dan Wang (2016) Social Segregation in Hong Kong's Schools: 2000–2012, Chinese Sociological Review, 48:3, 248-270

To link to this article: http://dx.doi.org/10.1080/21620555.2016.1166340

Published online: 25 May 2016.

Submit your article to this journal

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
http://www.tandfonline.com/action/journalInformation?journalCode=mcsa20

Download by: [University of Sussex Library] **Date:** 25 May 2016, At: 19:52

Chinese Sociological Review, 48(3): 248–270, 2016
Copyright © Taylor & Francis Group, LLC
ISSN 2162-0555 print / 2162-0563 online
DOI: 10.1080/21620555.2016.1166340

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

Social Segregation in Hong Kong's Schools: 2000–2012

Yisu Zhou, *Faculty of Education, University of Macau, Taipa, Macao SAR*
Tianji Cai, *Department of Sociology, University of Macau, Taipa, Macao SAR*
Dan Wang, *Division of Policy, Administration and Social Sciences Education, Faculty of Education, University of Hong Kong, Pok Fu Lam, Hong Kong SAR*

Abstract: *This article examines the distribution of students from various socioeconomic backgrounds across secondary schools in Hong Kong. Using data from the Program for International Student Assessment over the past decade, we computed axionomical indices of social segregation for 15-year-old students. The degree of social segregation in Hong Kong is compared with three other Chinese societies: Macau, Taiwan, and Shanghai. We found that social segregation steadily increased during 2000–2012. Hong Kong has the highest level of social segregation in schools among the four regions. The degree of segregation remains relatively stable among the top and bottom student socioeconomic status (SES) groups while substantially spread out among middle SES groups. Two factors were found to drive social segregation: gentrification of elite private schools, and the creation of semi-private schools for the upper middle class. Our findings suggest increasingly dividing education experiences among Hong Kong's youth.*

Social segregation in schools is defined as the uneven distribution of student socioeconomic status (SES) groups across schools. It has long been identified as a significant factor contributing to education inequality between

Address correspondence to Yisu Zhou, Faculty of Education (E33), University of Macau, Room 3022, Av. da Universidade, Taipa, Macao SAR. E-mail: zhouyisu@umac.mo
Color versions of one or more of the figures in the article can be found online at www.tandfonline.com/mcsa.

children of different classes. Jean Anyon's classic study in five socially segregated primary schools in the United States revealed shocking disparities in curriculum, pedagogy, and social control of the students among the schools. Social segregation among schools leads to the reproduction of social classes through the educational process (Anyon 1980; Peterson 2007). In contrast, a socially mixed learning environment is beneficial in narrowing the achievement gaps (Chudgar, Luschei, and Zhou 2013). As early as the 1960s, the Coleman Study in the United States discovered that an effective measure to reduce the achievement gap between middle-class white and lower-class African American students was to allow them to intermingle in the classroom and let them sit next to each other (Coleman 1966). Coleman's finding is corroborated by recent scholarship, which discovers that the production of cognitive skills in schools not only depends on the child's own ability and effort, but also on the peers with whom the child interacts (Hanushek et al. 2003). When schools are segregated by socioeconomic status, youth from under-privileged families not only lose access to quality teachers and learning resources, but also lose the opportunity to be exposed to role models and social networks conducive for learning (Wilson 2012). Therefore, social segregation between schools is a salient indicator of inequality in an educational system.

The degree of social segregation between schools is prone to the influence of social and educational policies. For instance, zoning policies in major U.S. cities prohibit poor African Americans from moving out of economically troubled inner cities, and maintained a separation between the affluent suburban and impoverished urban schools (Grant 2009). However, in places like Raleigh, North Carolina, where policies enable desegregation by limiting every school's intake of students receiving free lunch (a proxy for poverty) to a maximum of 30 percent, in addition to offering two-way busing to break the residential segregation between the well-off suburbs and the poverty-stricken inner city, the achievement gaps have been successfully reduced, and an improvement in students' learning outcomes has been observed across the board (Grant 2009). Apparently, educational policies can play a vital role in orienting the school system toward or away from segregation.

However, in the past decades, educational policies have been heavily shaped by a neoliberal mentality that champions school choice and market competition (Lall 2012). Since the1980s, school choice reforms have gradually spread from the developed countries, such as the United States (Chubb and Moe 1990), England, Australia, The Netherlands, New Zealand, Sweden (OECD 1994), Israel (Shapira and Cookson 1997), Spain (Escardíbul and Villarroya 2009), to other parts of the world like Chile, Colombia, and Bangladesh (Adnett and Davies 2002). Hong Kong also embraced this idea (Tse 2008). The impact of choice policies on educational equality and social segregation has become a topic of heated debates.

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

Some choice proponents contend that school choice would liberate low-SES children and families from residential segregation and failing schools, and provide them with access to high-performing schools (Coons and Sugarman 1978; Archbald 2000, 2004). Thus, choice plans tend to drive toward equality and integration. Other choice advocates believe in the transformative power of market competition. Once parents vote with their feet, the jungle rule will drive out the under-performing schools, and therefore improve the educational efficiency in general (Chubb and Moe 1990; Hoxby 2003). Nevertheless, the critics of choice reforms refute these arguments with reputed evidence that school choice policies worsen rather than improve social segregation and educational inequality (Ball, Bowe, and Gewirtz 1996; Goldstein and Noden 2003; Bosetti 2004; Gabay-Egozi 2015).

This article examines the changing patterns of social segregation in Hong Kong's school system between 2000 and 2012. The Hong Kong government launched a pro-choice reform, the Direct Subsidy Scheme (DSS), approximately twenty years ago. The relatively long time span has made it possible to observe the policy impact, if any, on social segregation in schools. Much of the literature on school choice and social segregation concentrates on school systems in Western countries, with little empirical evidence from other regions in the world. As the neoliberal choice reforms penetrate other parts of the world, the Hong Kong case is significant to understand the changing patterns of social segregation in non-Western school systems and will provide important empirical evidence from an Asian society to the debate on school choice and social segregation.

This study sets out to answer two questions: (1) did social segregation in Hong Kong's schools increase after the introduction of the DSS scheme? And (2) did the introduction of DSS alter school composition based on socioeconomic status and to what degree? We utilize the Program for International Student Assessment (PISA) data from 2000 to 2012 to calculate social segregation in public, DSS, and private/international schools. The Hong Kong case is also compared with three other Chinese regions (Shanghai, Taiwan, and Macau) in PISA. It is the first empirical effort to describe the changing landscape of social segregation in Hong Kong's educational system.

Literature Review

School Choice and Social Segregation

School choice reform has been a global phenomenon since the 1980s (Whitty, Power, and Halpin 1998). As a reaction against the so-called public monopoly education, choice reforms claim to empower the parents by giving them the freedom to choose where to send their children for schooling (Ball 1993). Specific choice policies take varied forms including voucher, semi-private

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

schools such as charter schools in the United States, independent schools in Singapore (Chan and Tan 2008), concerted private schools in Spain (Escardíbul and Villarroya 2009), and inter-district or intra-district choices within the public system (Lauen 2007). Public funding will follow parents' choices of schools, and public schools are urged to compete with private and semi-private schools for students and government funds. The school choice programs have stimulated heated debates in recent decades about their impact on school segregation and educational equity.

Proponents of choice reforms ground their support on two arguments: the liberation thesis and the competition thesis (Lauen 2007). The liberation thesis advocates school choice as a way to "liberate" the disadvantaged students and families from their dysfunctional neighborhood schools that are increasingly unable to serve their needs (Coons and Sugarman 1978; Archbald 2004; Viteritti 2010). Middle-class families with higher income and educational levels secure their access to premium schools by exercising "choice through mortgage," while poor families are locked in run-down neighborhoods with failing schools (Gabay-Egozi 2015). The choice reform allows the working-class children to be transferred to better schools outside their catchment areas, thus potentially reducing class (and racial) segregation and enhancing educational equity.

The competition thesis places faith in the market to "lift all boats" for all students. The famous Chubb and Moe (1990) treatise maintains that switching poor students from low-performing schools to high-performing schools will provide a mechanism of natural selection through competition, which will force the failing schools either to transform themselves or to be closed down. The public school system, particularly, will be compelled to improve its productivity when facing the threats of voucher, charter schools, and private schools (Hoxby 2003). Although the competition thesis does not promise integration, it paints a bright future for education in general, in which students from poor and working class families are better served than in the system without choice.

However, the optimistic promises of choice plans are confronted by grave empirical challenges. Evidence supporting the positive effects of choice on integration and equity is scarce and far from conclusive. Gorard (1999) anticipates segregation to increase after the first several years of the choice policy implementation; yet, the overall level of segregation will eventually decrease. Gorard (1999) argues that working-class parents may be disadvantaged in knowledge and social or cultural capital required to exercise choice in the early stages of deciding on an ideal school, but eventually information regarding educational options diffuse to every family with the establishment of the full-scale market. In essence, these parents are lacking in both the ability to access reliable materials and the social connections needed to effectively inform their decisions and navigate the academic options for their children. These claims are backed by empirical findings from data from

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

England and Wales (Gorard and Fitz 1998a, 1998b; Gorard, Fitz, and Taylor 2001). However, these findings encounter criticisms and counter-evidence that indicate a steady rise of segregation in the United Kingdom in the 1990s (Noden 2000; Goldstein and Noden 2003). In fact, because the degree to which families have exercised their right to choose their children's schools has been so inconsistent, the British government has found it necessary to provide "choice advisers" to working-class parents to coach them to think and choose like the middle-class parents (Exley 2013).

Indeed, more empirical observations confirm the argument that choice policies have exacerbated social segregation. Goldstein and Noden (2003) using a multilevel approach found a consistent increase in segregation between schools in areas with selective secondary schools in the United Kingdom. Charter schools in the United States are found to contribute to, rather than mitigate, school segregation by class and race (Renzulli and Evans 2005; Cobb and Glass 2009). Middle-class parents are more savvy than lower-class parents in making educational choices. Middle-class parents actively avoid schools with a high proportion of disadvantaged intake. Students from affluent families are less likely to have school peers from low-income families (Altonji and Mansfield 2011). Burgeoning private and charter schools enabled wealthy families one more way to flee from integrated schools, leaving behind the low-income families to become concentrated in struggling schools (Reardon and Bischoff 2011). In contrast, families in poverty have little chance to enroll in high-performing schools due to the constraints of transportation, information, social network, and their children's academic history (Goldring and Hausman 1999; Bell 2009; Gabay-Egozi 2015). The middle classes benefit while the lower classes have to make do with the leftovers (Reay, David, and Ball 2001; Leathwood 2004). The choice policies that promise more integration and equality turn out to aggravate the social segregation among schools and reinforce and reproduce the middle-class advantages (Ball, Bowe, and Gewirtz 1996; Bosetti 2004).

Recent scholarship argues that the gap between the promises made and the working realities of school choice plans is due to the failures of policy design and implementation. Regulated school choice programs that are need-based and that closely oversee student assignment and selection into schools have the potential to reduce segregation (Cobb and Glass 2009; Koedel et al. 2009; Viteritti 2010). The liberation thesis presumes that the choice policies should target children and families from disadvantaged backgrounds with equity in mind. However, the competition thesis champions open enrollment and unrestricted choice for all, which effectively strengthens the advantages of families with higher socioeconomic status, thus exacerbating school segregation (Fiske and Ladd 2000; Cobb and Glass 2009; Koedel et al. 2009). This literature highlights the importance of policy design when examining the integrating or segregating effects of school choice programs. Neither embracement nor condemnation in general will do justice to the potential social

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

impact of school choice plans. This study, therefore, will illustrate the segregating effect of a school choice program in Hong Kong, namely the DSS, in light of the unregulated nature of the policy.

The DSS Schools in Hong Kong

Hong Kong is a global city (Chiu and Lui 2009) with alarming income inequalities. The Gini coefficient of Hong Kong climbed from .43 in the 1970s to .537 in 2011 (Government of the Hong Kong SAR 2012). In 2013, the top 1.4 percent of all employees received monthly salaries over HK$100,000 (US$12,820) while the median salary was HK$13,000 (US $1,666) (Census and Statistics Department 2014, 50). As early as 2001, the richest 10 percent of families possessed 41.2 percent of the total household income, while the bottom 10 percent had only .9 percent (Zhao, Zhang, and Sit 2004). In 2009, one out of four children up to 14 years old lived in households either without income or with income less than half of the average government welfare payment (around HK$1,500 or US$192) (Financial Secretary's Office 2010, 15–16).

The school system in Hong Kong consists of traditional public schools (TPS), international schools, local private schools, and newly emerging DSS schools. The traditional public school system includes government and aided schools, where students enroll with free tuition. TPS are fully funded by the government and free of charge to all students. The DSS schools, private, and international schools charge annual fees ranging from approximately HK$30,000 to HK$183,000 per year, an amount far beyond the reach of most households.

The DSS policy aimed to break the state monopoly of schooling, provide parents with more choices, and boost competition in the educational market (Yung 2006). The program was first launched in 1991 with the government's intention to strengthen the private school sector (Tse 2008). However, for various reasons, the plan failed to attract much interest and the number of DSS schools stagnated in the 1990s (Tse 2008). In 1997, there were only twelve DSS schools, most of which were private or international schools (Chan and Tan 2008). The response of the public schools, aided or government, remained cold.

In 1999, the Hong Kong government dramatically revamped the DSS in order to "foster the development of a vibrant DSS school sector to inject diversity and choice in the education system" (Education Department 2001, 1). Under this revised scheme, the DSS schools continue to enjoy tremendous autonomy in staff employment, student admission, school management, curriculum, and median of instruction, but the tuition fee cap was significantly raised. To attract schools to join the DSS sector, the new scheme permits the DSS schools to raise the tuition fee up to two and one-thirds of the average unit cost of an aided school while also

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

remaining entitled to receive government subsidies (Education Department 2001). This means that, even though a DSS school has not yet charged the maximum amount, in principle, these schools can enjoy over three times the amount of per student funding vis-à-vis a public school. On top of that, the government encourages elite public schools to be converted to DSS schools by granting them extra subsidies on infrastructure and equipment (Chan and Tan 2008).

Indeed, after the policy revision, the number of DSS schools exhibited a steady increase (Chan and Tan 2008). By September 2014, the figure climbed to 73 schools in total, including 52 secondary, 12 primary, and 9 secondary-cum-primary schools (Education Bureau 2015). The school fees range from HK$5,000 to as much as HK$60,000 per year. Although less than twenty schools were converted from aided schools, almost all of them are traditionally elite schools in the public system, including the most reputable schools like St. Paul Co-Educational College and Primary School, Diocesan Girls' School, and Diocesan Boys' School. These schools used to be open to all classes, but now have become the most expensive ones in the DSS sector (Yung 2006).

The DSS program is an unregulated choice plan, exercising little oversight over student assignment or the balance of student composition. By converting into DSS schools, these former elite public schools break free from the constraints of school districts as well as enjoy autonomy in student admission. DSS school enrollment is not limited by the school district they are located in. If they remained in the public system, 70 percent of their students would be assigned by the government's central allocation system. Many schools openly claim they practice meritocratic, need-blind admission policy. However, the government mandates that DSS schools have to reserve a minimum of 10 percent of total tuition income for fee remissions for poor students. In addition, if the tuition charge exceeds two-thirds of the per capita aided student funding, the school is obliged to set aside half of the proceeds above that amount as scholarship or financial assistance for needy students (Education Department 2001). The provision of financial means may offset part of the segregating tendency.

The introduction of DSS makes Hong Kong's education system stand out among other Chinese societies. Although private and semi-private schools existed in places such as Taiwan, Macau, and Shanghai, they simply have not reached the critical mass compared to Hong Kong. In 2012, DSS enrolls about 10 percent of total students. Taiwan launched its charter school policy as early as 1999. By 2011 there were only five charter schools in Taiwan (Lee 2011). Altogether, the private sector comprised only 1.4 percent and 2.2 percent, respectively of the elementary and junior secondary schools, but 43 percent of the high schools in Taiwan (World Education News and Reviews 2010), and therefore, is too small in lower grades and too big at the high school level to have a system-wide impact

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

on segregation. In comparison, school choice in Shanghai exhibits the unique feature in that parents mostly pursue elite "key" schools within the public sector because the private sector is still in its early stage (Wu 2012). Still enrollment in Shanghai is largely confined within school districts, which makes "back-door" school choice a small-scale phenomenon. Macau comes closest to the Hong Kong system. Even though majority of Macau schools are private, the government has attracted most of them (54 out of 64 schools in 2014) to join the Free Education Network. These so-called in-net schools essentially gave up their autonomy, including regulated admission process, in exchange for public resources. The remaining ten private schools outside the Network are given significant amount of subsidies (Education and Youth Affairs Bureau 2014). Apparently none of the three regions has exercised school choice policies as aggressive and unregulated as those in Hong Kong.

The DSS program has long been viewed by Hong Kong's public as a newly emerged enclave for children from wealthy and elite families. However, there is little systematic empirical evidence to testify this hypothesis. If DSS schools focus exclusively on recruiting high SES students, what is the implication for schools that lose students?

Data and Method

Data

We use data from PISA. PISA is one of the largest international student assessment programs conducted by the Organization for Economic Cooperation and Development (OECD) on a three-year basis. The latest iteration of PISA includes more than seventy global economies. Hong Kong was among the first adopters of PISA since 2000, thus we have five available waves of PISA data spanning from 2000 to 2012. One feature of PISA is that four Chinese societies participated. The remaining three: Macau (2003–2012), Taiwan (2006–2012), and Shanghai (2009–2012), each operates under different socio-political regimes. PISA's standardized measurement approach allows us to use these societies as a comparison to Hong Kong.

PISA's target population is 15 year olds approaching the end of the compulsory education cycle. PISA uses a two-stage stratified sampling scheme. In the first stage, schools are selected. In the second stage, at least thirty-five students are randomly selected within each school. Students are administered a background questionnaire and a battery of tests on the three test subjects of reading, mathematics, and science. School-level information is solicited from the school principal by completing a school survey. The sample is described in Table A1. The PISA survey has a high response rate, with each cycle averaging above 90 percent for responses.

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

Variables

Socioeconomic Status

The index variable is created by OECD by combining three aspects that measure the social standing of the student: highest level of parental education; highest level of parental occupational status; and an index of family possessions including common items such as books, computers, and regional-specific items.[1] This student self-reported variable is one of the most commonly used in PISA-related studies. In each region, we divide the index into ten groups, representing the highest SES decile to the bottom SES decile.

Household Income

This categorical variable is taken from the optional parental questionnaire, which provides additional information about socioeconomic status. Since only the parents of students in Hong Kong and Macau opted in, we have used the data from these two regions. The income level is measured on a six-category classification. For each region, the relative categorization is decided based on each region's median income. The highest category is chosen if the parent-reported data is 1.5 times the regional median. The lowest category is chosen if the parent-reported data is below 0.5 times the regional median (OECD 2009).

School Type

School type measures whether the school is a public or private school. The information is taken from a mandatory school questionnaire answered by the principal. Additionally, for Hong Kong, a secondary classification scheme is also reported in the technical literature. It uses a four-category classification: government school, aided school, international/private school, and DSS schools. The former two are considered traditional public schools, while the latter two are private or semi-private institutions (Tse 2008).

Measure of Social Segregation

To measure the degree of social segregation in schools in a multigroup setting, we use a descriptive index: the Entropy Information H index (Reardon and Firebaugh 2002; Mora and Ruiz-Castillo 2011). The H index is defined as such:

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

$$H = \frac{M}{E(P_g)} = \sum_{n=1}^{N} p_n \left[\frac{E(P_g) - E(P_{g|n})}{E(P_g)} \right] \tag{1}$$

The H index is closely related to the idea of entropy from information theory (Theil 1967, 1971), where $E(\cdot)$ is the entropy function:

$$E(P_q) = -\sum_{q=1}^{Q} p_q log(p_q) = \sum_{q=1}^{Q} p_q log(\frac{1}{p_q})$$

The entropy function reflects the amount of information revealed by observing the actual value of a student's membership of a particular socio-economic group. Mathematically, the entropy is the opposite of the logarithm of its likelihood, $-log(p_q) = log(1/p_q)$, where p_q is the proportion of students in the area who belong to the q th group, and $p_{g|n}$ is the proportion of students in the qth group in school type n. Therefore, the entropy is a measure of diversity.

The denominator in Equation (1), M, is also called the Mutual Information index. Conceptually, the M index is constructed as the weighted average of change in entropy. It reflects the difference between the maximum possible entropy (i.e., information about the distribution of students from various SES groups in the area), minus the observed entropy in schools (i.e., information about the distribution of students from various SES groups in one particular school), weighted by the relative size of the school:

$$M = \sum_{\substack{\text{schools } n \\ \text{in area}}} \left(\frac{\text{Students in school } n}{\text{Students in the area}} \right) \left(\begin{array}{c} \text{Entropy of area's SES distribution} \\ \textit{minus} \\ \text{Entropy of school } n\text{'s SES distribution} \end{array} \right)$$

But since M does not have an easily interpretable metric, we normalize it in Equation (1) to be bounded between 0 and 1, where 0 indicates no segregation exists, and 1 when there is total segregation. The H index can be interpreted as one minus the ratio of average within unit diversity to the diversity of total population.

The appeal of entropy-based segregation indices over other traditional segregation indices such as dissimilarity (Duncan and Duncan 1955) or exposure is that it satisfies several other criteria (Frankel and Volij 2011; Mora and Ruiz-Castillo 2011). Secondly, we can compute percentage of contribution to overall segregation by the sub-unit of analysis:

$$\%\ contribution\ for\ school\ type\ n = \frac{P_n(E(P_g) - E(P_{g|n}))}{M} \tag{2}$$

The additive decomposition feature allows us to evaluate the percentage contribution of segregation by various subgroups.

Lastly, since PISA is a survey rather than a census of all schools in these regions, we follow Jenkins, Micklewright, and Schnepf (2008) to use

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

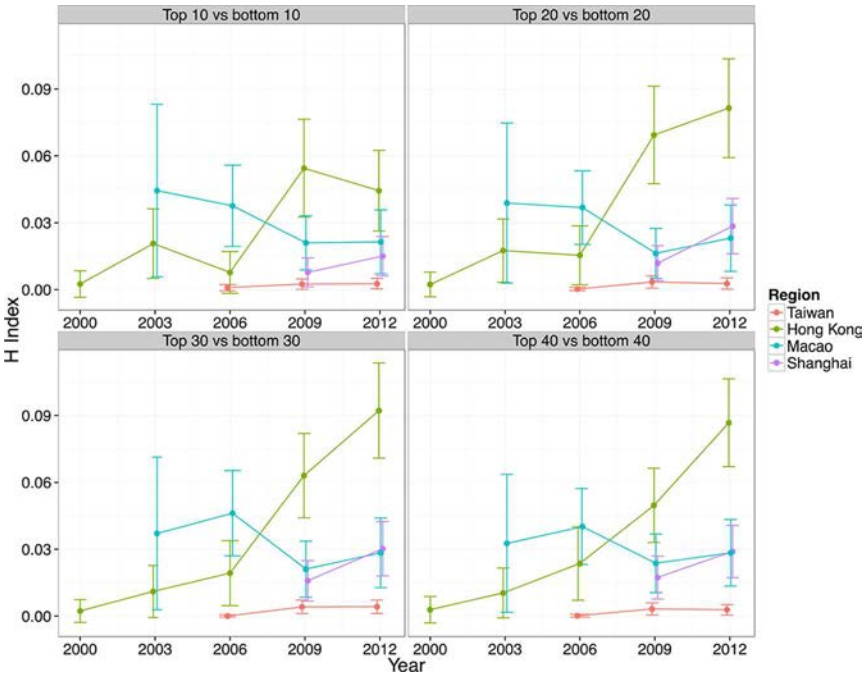
bootstrap methods with 1,000 replication to compute empirical standard errors of the segregation indices.

Analysis

We first present the overall segregation among various socioeconomic groups in the four Chinese regions using all available PISA data. We compute the *M* and *H* index based on ten SES categories. Figure 1 shows the *H* index for each region using public/private school data from PISA. The top left shows the segregation index among the highest 10 percent and lowest 10 percent SES students. The top right graph shows the segregation index among highest 20 percent and lowest 20 percent SES students. The two graphs on the bottom use similar methods for grouping. We combine all available PISA data since 2000 and use three Chinese societies as comparison.

Several findings can be inferred from Figure 1. First, the overall pattern of segregation is strikingly similar regardless of the grouping methods we chose. Social segregation increased steadily from 2000–2012, and the rate of segregation seems accelerated since 2006 in Hong Kong.

Figure 1. **Social segregation in public and private schools among socioeconomic groups, 2000–2012.**



Downloaded by [University of Sussex Library] at 19:52 25 May 2016

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

Secondly, the major change of pattern of social segregation seems to be coming from the middle tier of SES groups. For instance, when we consider the top 30 percent and bottom 30 percent of students, the level of segregation is much larger than if we were only comparing the top 10 percent and bottom 10 percent of students. Thirdly, segregation did not only occur within one particular segment of the student population, but rather was a prevalent issue among a wide group of students. This is implied from the fact that levels of segregation between the top 40 percent and bottom 40 percent are almost the same as between the top 20 percent and bottom 20 percent. Traditionally, access to private schooling, typically characterized by high tuition and selective admission based on students' family background, is regarded as a privilege enjoyed by a handful of elite families. Under this scenario, segregation between the top and bottom SES groups should be the highest, as private schools exclusively enroll high SES students, and the student mix in public schools constitute more low SES students. Figure 1 suggests that the movement toward a more segregated school system occurred not at the extreme end of the socio-economic distribution. Comparatively, segregation among students from middle SES groups increased at a much faster pace. This means that at a societal level, public schools and private schools are increasingly segregated in Hong Kong.

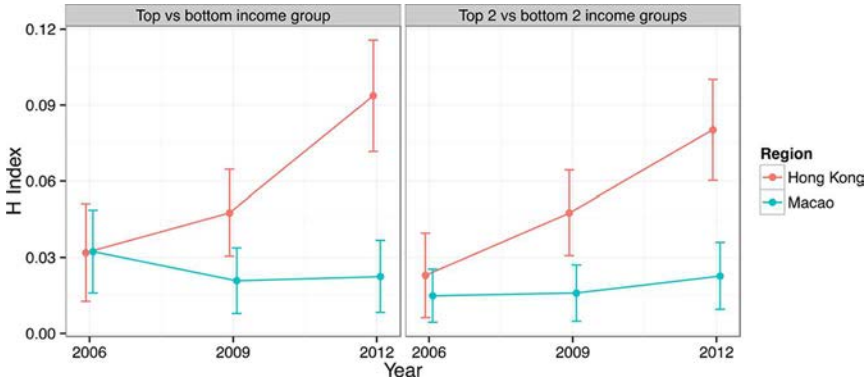
Thirdly, the trend of segregation seems accelerated in the second half of the first decade of the twenty-first century. The majority of change in segregation occurs post-2006 in Hong Kong, where we observe a dramatically increased level of social segregation. Fourthly, from a comparative perspective, the level of segregation is the highest in Hong Kong amongst the four available Chinese societies in the PISA study. With available data, social segregation in Hong Kong is comparable in 2003 with that of Macau. But the level of segregation actually declined slightly in Macau in the subsequent years, and remained stable across the different SES groups. Yet, we witness a sharp increase of segregation in Hong Kong during the same period of time. In the remaining two regions, Shanghai and Taiwan, the level of segregation is also only a fraction of what is in Hong Kong.

Due to the potential for 15-year-old students to misrepresent their family's SES standings on the self-reported questionnaire, we also use the parental feedback on household income as an alternative measure of SES. We present our estimates in Figure 2.

The level of segregation and change in trends is very similar to earlier results using student SES as a measure of social groups. Again, we find accelerated growth in segregation from 2006 onward in Hong Kong, whereas Macau remains fairly stable. In 2012, segregation in Hong Kong is more than three times that of Macau.

A major reform in the education sector in Hong Kong is the (re)rolling out of the DSS in the early 2000s. As documented in Luk (1990) and Tse

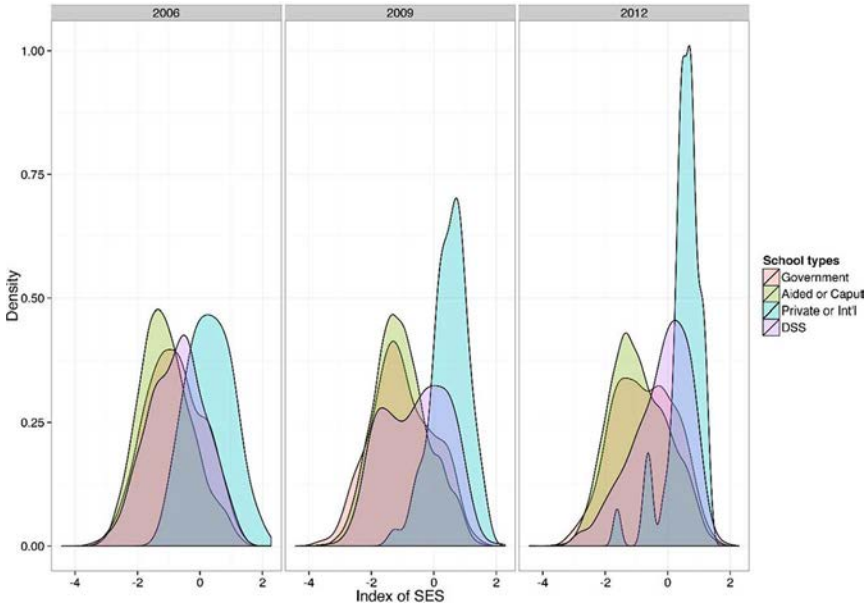
Figure 2. **Segregation based on family income in public and private schools, 2000–2012.**



(2008), the main feature of DSS is increased autonomy in nearly every aspect of school management while still being funded using public resources. We show the student SES distribution in each of the four types of schools in Hong Kong in Figure 3.[2]

Figure 3 suggests two important things. First of all, private international schools became increasingly "gentrified" in the sense that its student

Figure 3. **Student SES distribution in Hong Kong, 2006–2012.**



Downloaded by [University of Sussex Library] at 19:52 25 May 2016

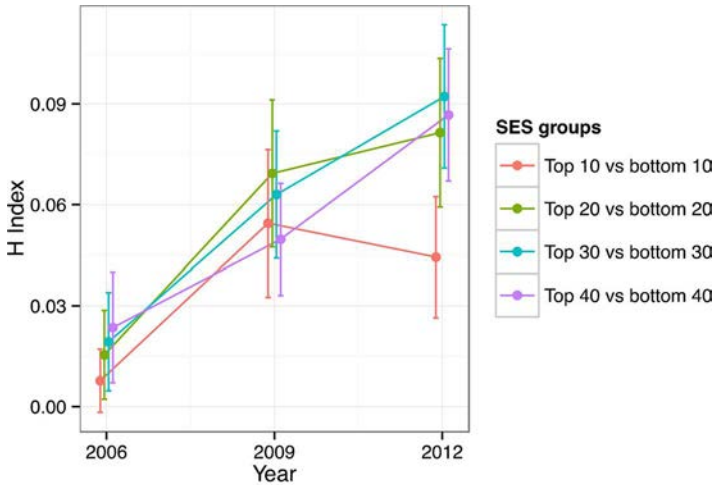
population was extremely homogenous and inclusive of only very high-SES students. Secondly, the DSS schools, as having more autonomy in admission and other aspects of management, gradually replaced its student body with more upper-middle class students, as marked by a right-shift of the density curve as well as smaller variation. While there is still a significant proportion of DSS students coming from disadvantaged family backgrounds, gradually their DSS schoolmates are coming from high-SES families. On the other hand, the change of student SES distribution in DSS and international schools is countered by a downward shift of student composition in aided schools.

Could change in student body translate into change in levels of segregation? We also calculate the *H* index based on a four-group categorization of schools, and present the estimates in Figure 4.

Again, the pattern is similar to previous estimates. The segregation does not seem to be driven by the separation between the most well-off students and the least; one can probably argue they are already miles apart. Instead, we see a broader increase in social segregation among the middle tiers of SES groups. This shows that the haves and have-nots students are becoming more and more separate.

To understand the source of segregation, we break down proportional contribution to social segregation by different types of schools using 2006–2012 data, which reveals a major change (Table 1). In 2006, private and international schools contributed half of the social segregation between the top 10 percent and bottom 10 percent of student groups, and 42 percent of segregation between the top 30 students and the bottom 30 students. This is no

Figure 4. **Social segregation among four school types in Hong Kong, 2006–2012.**



Note: estimation based on 4 school types

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

Table 1

Proportional Contribution to Social Segregation by School Types

	Types of schools			
	Government	Aided	Private or international	DSS
	2012			
Top 10 vs bottom 10	2%	14%	29%	54%
Top 20 vs bottom 20	0%	16%	29%	55%
Top 30 vs bottom 30	0%	15%	22%	63%
Top 40 vs bottom 40	0%	15%	19%	66%
	2009			
Top 10 vs bottom 10	3%	12%	62%	23%
Top 20 vs bottom 20	1%	11%	59%	28%
Top 30 vs bottom 30	1%	10%	58%	31%
Top 40 vs bottom 40	0%	8%	63%	29%
	2006			
Top 10 vs bottom 10	23%	13%	50%	14%
Top 20 vs bottom 20	19%	13%	44%	24%
Top 30 vs bottom 30	15%	13%	42%	30%
Top 40 vs bottom 40	13%	14%	34%	39%

surprise given the user-pay principle and high fee structure of private schools. However, six years later, international schools are no longer the largest contributor of segregation (this is not to be confused with the level of segregation), instead the DSS sector has become the largest contributor. International schools contribute less than 30 percent to the segregation among top and bottom SES groups, whereas, DSS schools contribute more than half to the segregation between different SES groups. This result is consistent no matter what grouping method we use (see the last column of the first panel of Table 1). Conversely, the relative contribution to social segregation decreased in government schools and remained the same in aided schools.

Discussion

Before we move into interpreting our findings, the limitation of this study needs to be addressed. This study provides the first systematic empirical assessment of the degree of social segregation in Hong Kong and three other Chinese societies. However, results from the current research need

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

to be treated with caution due to the limitation of PISA data. Since the PISA survey is self-reported by students, the degree to which 15 year olds can accurately describe their family's social standing is a debatable question. Though we corroborate student data with parental data and found similar patterns, further study is nonetheless needed. Secondly, though the sampling scheme in PISA ensures that students are randomly chosen within schools and schools are representative of each type, inevitably in each cycle PISA sampled different schools. This limitation constrains the comparability of the segregation index computed from each cycle. Thus, our results are only interpreted as segregation between school types instead of across individual schools since only school type designation remained identical across cycles. We also computed empirical standard errors of indices to show the range of error that could be introduced by the sampling process. Using census data provides the most accurate base for calculation of segregation, but to our knowledge, Hong Kong's education authority did not collect information on student socioeconomic standing. There is no second survey that collects comparable measures across different regions either. Our research aims to contribute to the current scholarship on social segregation despite the data limitations and imperfect measurements of evaluation.

In Western societies such as the United States, social segregation is historically entangled with racial and ethnic segregation. In recent decades however, when overall racial segregation recedes, social segregation exploded as income inequality took the main stage (Massey, Rothwell, and Domina 2009). Yet in Hong Kong, the scope of social segregation in schools is not very well understood. Our previous study (Zhou, Wong, and Li 2015) hinted at the possibility that recent marketization would lead to social segregation. In this article, we formally assess this issue from a medium-run time aspect.

The analysis shows that Hong Kong's educational system has the highest degree of social segregation among the four Chinese regions participating in the PISA survey. The degree of separation based on family background is three times higher in Hong Kong than Macau, Taiwan, and Shanghai. More alarmingly, judging by the latest data from 2012, there is no sign of stopping the trend of segregation in Hong Kong. Given that Taiwan and Shanghai have relatively lower degree of privatization and consequentially higher level of regulation on student admission at lower secondary level, it is not surprising that social segregation is lower in these two societies. Macau remains an interesting case for future study. Macau schools retain substantial autonomy in student admission and the level of social segregation across different SES groups remain fairly stable, higher than those of Taiwan and Shanghai but lower than that of Hong Kong. Unlike the upward trend of H-index in Hong Kong, there has not been much change in Macau.

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

Our evidence indicates that social segregation in Hong Kong's educational system substantially increased in the first decade of the twenty-first century. The Hong Kong case is consistent with the observations in other countries that unrestricted choice plans intensify school segregation by class (Fiske and Ladd 2000; Cobb and Glass 2009; Koedel et al. 2009). Students from different SES groups are increasingly attending different types of schools. As demonstrated in Figure 3, by 2012, schools in Hong Kong had been clearly divided into three tiers by students' socioeconomic status. The traditional distinction between the private sector (private and international schools) and the public sector (government and aided schools) persists. The highest SES students predominantly attend private and international schools while the public schools primarily serve working-class and lower-middle class children. However, a third track in-between the private and public sector has emerged thanks to the new school choice program, the DSS. The DSS schools experienced a significant transformation in the first decade of the new millennium. While in the early 2000s the DSS schools were comparable to the public schools in student composition, by the end of the decade they exhibited a disproportional concentration of students from upper-middle class families. As the result, DSS schools are the largest contributor of social segregation among Hong Kong's schools today. They exacerbate the educational inequality already existing between the public and the private/international schools.

Our analysis leads us to believe that this unregulated market-oriented strategy of education reform, by introducing a new form of semi-public institutions, which gives more unfettered choice to the parents, is clearly benefiting certain populations at the cost of others. It offers a new way out for affluent families from traditional public education. But the consequence of this new reform is highly separated schools where low-SES students are left in public institutions.

Moreover, the admission standards, which claim to be meritocratic, are in fact vague and opaque. Because the government prohibits DSS schools to administer written examinations for admission, the selection of a student, therefore, highly depends on the curated presentation of the child's portfolio and the interview process with the child and the parents; both of which work subtly in favor of parents with a higher level of education and children with middle-class manners. The family human capital and cultural capital are even more salient in the admission to the DSS primary schools since the variation in children's academic achievement is negligible at this early stage. In other words, meritocracy as the admission principle gives the DSS schools a convenient excuse to handpick the middle-class students they prefer. The specific mechanisms through which parents exercise these resources are worthy of further research.

This study identifies the source of the growth in segregation within the middle-class. While the wealthiest and the poorest children continue to live

and learn in social circles with little overlapping with others, the DSS program exacerbates social segregation by making a new division within middle-class families. The evidence suggests that middle-SES children, those who stand at the 60th and 70th percentile on the SES scale, are now attending separate schools from those children at the 40th or 30th percentile. This is a new phenomenon in Hong Kong as the division of middle-class families now extends to middle school years. We speculate that this phenomenon might echo the argument by Chiu and Lui (2009) that globalization has created a more nuanced stratification between the upper-middle class, such as the business, administrative, academic, and legal professionals, and the lower-middle class, like clerks, lower-level managers, and teachers. The increased separation between the children in these two groups may be a reflection of the expanding inequality within the middle-class in the larger Hong Kong society. The implications of such a division of the middle-class on the growing economic inequality, which in large part could be contributable to the variation in education, is alarming (Wong 2013). It could mean that the educational inequality for other marginalized groups, such as children of the new migrants (Zhang 2014; Xu and Wu 2015b) and female students from lower strata (Xu and Wu 2015a), is being intensified and solidified. Future research with comprehensive parental information is needed to testify this emerging trend of social stratification.

In conclusion, Hong Kong's case illustrates that the unregulated school choice plan based entirely on market competition ultimately works to the advantage of the upper and middle classes. The so-called parental choice is a privilege enjoyed only by the families with sufficient economic resources, social networks, and cultural capital. Such program has resulted in an increasingly segregated education system in comparison with other Chinese societies. School choice reforms without close government intervention to encourage social integration will intensify social segregation between schools, thus further aggravating, rather than reducing, educational inequality.

Acknowledgements

The feedback from two anonymous reviewers and the editor are highly appreciated. Tianji and Yisu share equal contribution to this article.

Funding

The authors are grateful for financial support from the University of Macau via two research grants, MYRG2015-00052-FED and MYRG2015-00002-FSS.

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

Notes

1. For details, see OECD (2014b).

2. The original social economic index is constructed in such a way that makes comparing across the Program for International Student Assessment (PISA) cycles not meaningful. According to the Organization for Economic Cooperation and Development's (OECD's) recommendation, using the common items that existed in all three waves of PISA data in 2006, 2009, and 2012, we rescale the student socioeconomic status (SES) variable to make it comparable across cycles. For details, see OECD (2014a, 258).

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

References

Adnett, N., and P. Davies. 2002. *Markets for Schooling: An Economic Analysis*. London: Routledge.

Altonji, J. G., and R. K. Mansfield. 2011. "The Role of Family, School, and Community Characteristics in Inequality in Education and Labor Market Outcomes." Chapter 16, pp. 339–358 in *Whither Opportunity?: Rising Inequality, Schools, and Children's Life Chances*, edited by G. J. Duncan and R. J. Murnane. New York: Russell Sage Foundation.

Anyon, J. 1980. "Social Class and the Hidden Curriculum of Work." *The Journal of Education* 162(1): 67–92. Retrieved from www.jstor.org/stable/42741976

Archbald, D. A. 2000. "School Choice and School Stratification: Shortcomings of the Stratification Critique and Recommendations for Theory and Research." *Educational Policy* 14(2): 214–40. doi:10.1177/0895904800014002002

——. 2004. "School Choice, Magnet Schools, and the Liberation Model: An Empirical Study." *Sociology of Education* 77(4): 283–310. doi:10.1177/003804070407700402

Ball, S. J. 1993. "Education Markets, Choice and Social Class: the market as a class strategy in the UK and the USA." *British Journal of Sociology of Education* 14(1): 3–19. doi:10.1080/0142569930140101

Ball, S. J., R. Bowe, and S. Gewirtz. 1996. "School Choice, Social Class and Distinction: The Realization of Social Advantage in Education." *Journal of Education Policy* 11(1): 89–112. doi:10.1080/0268093960110105

Bell, C. A. 2009. "All Choices Created Equal? The Role of Choice Sets in the Selection of Schools." *Peabody Journal of Education* 84(2): 191–208. doi:10.1080/01619560902810146

Bosetti, L. 2004. "Determinants of School Choice: Understanding How Parents Choose Elementary Schools in Alberta." *Journal of Education Policy* 19(4): 387–405. doi:10.1080/0268093042000227465

Census and Statistics Department. 2014. *Hong Kong Annual Digest of Statistics 2014*. Hong Kong: Government of the Hong Kong SAR.

Chan, D., and J. Tan. 2008. "Privatization and the Rise of Direct Subsidy Scheme Schools and Independent Schools in Hong Kong and Singapore." *International Journal of Educational Management* 22(6): 464–87. doi:10.1108/09513540810895417

Chiu, S., and T. L. Lui. 2009. Hong Kong: Becoming a Chinese Global City. London: Routledge.

Chubb, J. E., and T. M. Moe. 1990. *Politics, Markets and America's Schools*. Washington, DC: The Brookings Institution.

Chudgar, A., T. F. Luschei, and Y. Zhou. 2013. "Science and Mathematics Achievement and the Importance of Classroom Composition: Multicountry

Analysis Using TIMSS 2007." *American Journal of Education* 119(2): 295–316. doi:10.1086/668764
Cobb, C. D., and G. V. Glass. 2009. "School Choice in a Post-Desegregation World." *Peabody Journal of Education* 84(2): 262–78. doi:10.1080/01619560902810187
Coleman, J. S. 1966. *Equality of Educational Opportunity*. Washington, DC: National Center for Educational Statistics, U.S. Department of Education.
Coons, J. E., and S. D. Sugarman. 1978. *Education by Choice: The Case for Family Control*. Berkeley and Los Angeles: University of California Press.
Duncan, O. D., and B. Duncan. 1955. "A Methodological Analysis of Segregation Indexes." *American Sociological Review* 20(2): 210–17. doi:10.2307/2088328
Education and Youth Affairs Bureau. 2014. "Overview of Educational Statistics 2014–2015." Retrieved from www.dsej.gov.mo/~webdsej/www/statisti/2014/index.html?timeis=Wed%20Dec%2030%2015:10:01%20GMT+08:00%202015&&
Education Bureau. 2015. *General Information on DSS*. Education Bureau. Retrieved from www.edb.gov.hk/en/edu-system/primary-secondary/applicable-to-primary-secondary/direct-subsidy-scheme/index/info-sch.html
Education Department. 2001. *Circular Memorandum no. 210/2001: Direct Subsidy Scheme (Tech. Rep.)*. Education Department. Retrieved from www.edb.gov.hk/attachment/en/edu-system/primary-secondary/applicable-to-primary-secondary/direct-subsidy-scheme/index/adm%20cir210-2001-e.pdf
Escardíbul, J., and A. Villarroya. 2009. "The Inequalities in School Choice in Spain in Accordance to PISA Data." *Journal of Education Policy* 24(6): 673–96. doi:10.1080/02680930903131259
Exley, S. 2013. "Making Working-Class Parents Think More Like Middle-Class Parents: Choice Advisers in English Education." *Journal of Education Policy* 28(1): 77–94. doi:10.1080/02680939.2012.689012
Financial Secretary's Office. 2010. *Indicators of Poverty: An Update of 2009*. Government of the Hong Kong SAR. Retrieved from www.lwb.gov.hk/eng/other_info/2009%20Poverty%20Indicators_eng.pdf
Fiske, E. B., and H. F. Ladd. 2000. *When Schools Compete: A Cautionary Tale*. Washington, DC: Brookings Institution Press.
Frankel, D. M., and O. Volij. 2011. "Measuring School Segregation." *Journal of Economic Theory* 146(1): 1–38. doi:10.1016/j.jet.2010.10.008
Gabay-Egozi, L. 2015. "School Choice in a Stratified Geography: Class, Geography, Otherness, and Moral Boundaries." *Journal of Education Policy* 31(1): 1–27. doi:10.1080/02680939.2015.1047410
Goldring, E. B., and C. S. Hausman. 1999. "Reasons for Parental Choice of Urban Schools." *Journal of Education Policy* 14(5): 469–90. doi:10.1080/026809399286161
Goldstein, H., and P. Noden. 2003. "Modelling Social Segregation." *Oxford Review of Education* 29(2): 225–37. doi:10.1080/0305498032000080693. Retrieved from www.jstor.org/stable/1050612
Gorard, S. 1999. "'Well. That About Wraps It Up for School Choice Research': A State of the Art Review." *School Leadership & Management* 19(1): 25–47. doi:10.1080/13632439969320
Gorard, S., and J. Fitz. 1998a. "The More Things Change ... The Missing Impact of Marketisation?" *British Journal of Sociology of Education* 19(3): 365–76. doi:10.1080/0142569980190306
——. 1998b. "Under Starters Orders: The Established Market, the Cardiff Study and the Smithfield Project." *International Studies in Sociology of Education* 8(3): 299–316. doi:10.1080/0962021980020030
Gorard, S., J. Fitz, and C. Taylor. 2001. "School Choice Impacts: What Do We Know?" *Educational Researcher* 30(7): 18–23. doi:10.3102/0013189x030007018

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

Government of the Hong Kong SAR. 2012. *Half-Yearly Economic Report 2012*. Government of the Hong Kong SAR. Retrieved from www.hkeconomy.gov.hk/en/pdf/box-12q2-5-2.pdf

Grant, G. 2009. *Hope and Despair in the American City: Why There Are No Bad Schools in Raleigh*. Cambridge, MA and London, England: Harvard University Press.

Hanushek, E. A., J. F. Kain, J. M. Markman, and S. G. Rivkin. 2003. "Does Peer Ability Affect Student Achievement." *Journal of Applied Econometrics* 18: 527–44. doi:10.1002/jae.741

Hoxby, C. M. 2003. "School Choice and School Competition: Evidence from the United States." *Swedish Economic Policy Review* 10(2): 9–65.

Jenkins, S. P., J. Micklewright, and S. V. Schnepf. 2008. "Social Segregation in Secondary Schools: How Does England Compare with Other Countries?" *Oxford Review of Education* 34(1): 21–37. doi:10.1080/03054980701542039

Koedel, C., J. R. Betts, L. A. Rice, and A. C. Zau. 2009. "The Integrating and Segregating Effects of School Choice." *Peabody Journal of Education* 84(2): 110–29. doi:10.1080/01619560902810096

Lall, M. 2012. *Policy, Discourse and Rhetoric: How New Labour Challenged Social Justice and Democracy*, vol. 52. Rotterdam and Boston: Springer Science & Business Media.

Lauen, D. 2007. "False Promises: The School Choice Provisions in No Child Left Behind." Chapter 25, pp. 461–486 in *Sociology of Education: A Critical Reader*, edited by A. R. Sadovnik 2nd ed. New York and London: Routledge.

Leathwood, C. 2004. "A Critique of Institutional Inequalities in Higher Education: (Or an Alternative to Hypocrisy for Higher Educational Policy)." *Theory and Research in Education* 2(1): 31–48. doi:10.1177/1477878504040576

Lee, C. -L. 2011. "A Comparative Study on the Private Management of Public Schools for Compulsory Education among Taiwan and Spain." Master thesis, National Chi Nan University, Taiwan.

Luk, B. H. 1990. "Privatization and Meritocracy: The Real Issue of the Direct Subsidy Scheme." Pp. 47–55 in *Differences and Identities: Educational Argument in Late Twentieth Century Hong Kong*, edited by A. Sweeting. Faculty of Education, University of Hong Kong.

Massey, D. S., J. Rothwell, and T. Domina. 2009. "The Changing Bases of Segregation in the United States." *The ANNALS of the American Academy of Political and Social Science* 626(1): 74–90. doi:10.1177/0002716209343558

Mora, R., and J. Ruiz-Castillo. 2011. "Entropy-Based Segregation Indices." *Sociological Methodology* 41(1): 159–94. doi:10.1111/j.1467-9531.2011.01237.x

Noden, P. 2000. "Rediscovering the Impact of Marketisation: Dimensions of Social Segregation in England's Secondary Schools, 1994–1999." *British Journal of Sociology of Education* 21(3): 371–90. doi:10.1080/713655353

OECD. 1994. *School, a Matter of Choice*. Paris, France: OECD Publishing.

OECD. 2009. *PISA 2006 Technical Report.* Paris, France: OECD Publishing. doi:10.1787/9789264048096-en

——. 2014a. *PISA 2012 Results: What Students Know and Can Do – Student Performance in Mathematics, Reading and Science*, vol. 1, revised edition. Paris, France: OECD Publishing. doi:10.1787/9789264201118-en

——. 2014b. *PISA 2012 Technical Report.* OECD Publishing. Retrieved from www.oecd.org/pisa/pisaproducts/pisa2012technicalreport.htm

Peterson, M. W. 2007. "The Study of Colleges and Universities as Organizations." Pp. 147–84 in *Sociology of Higher Education: Contributions and Their Contexts*, edited by P. J. Gumport. Baltimore, MD: Johns Hopkins University Press.

Reardon, S. F., and K. Bischoff. 2011. "Income Inequality and Income Segregation." *American Journal of Sociology* 116(4): 1092–153. doi:10.1086/657114

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

Reardon, S. F., and G. Firebaugh. 2002. "Measures of Multigroup Segregation." *Sociological Methodology* 32(1): 33–67. doi:10.1111/1467-9531.00110
Reay, D., M. David, and S. J. Ball. 2001. "Making a Difference?: Institutional Habituses and Higher Education Choice." *Sociological Research Online* 5(4). doi:10.5153/sro.548. Retrieved from www.socresonline.org.uk/5/4/reay.html
Renzulli, L. A., and L. Evans. 2005. "School Choice, Charter Schools, and White Flight." *Social Problems* 52(3): 398–418. doi:10.1525/sp.2005.52.3.398
Shapira, R., and P. Cookson. 1997. *Autonomy and Choice in Context: An International Perspective*. Oxford: Pergamon.
Theil, H. 1967. *Economics and Information Theory*. Amsterdam, The Netherlands: North Holland.
Theil, H. 1971. *Principles of Econometrics*. New York: Wiley.
Tse, T. K. 2008. "Choices for Whom?: The Rhetoric and Reality of the Direct Subsidy Scheme in Hong Kong (1988–2006)." *Education and Urban Society* 40(5): 628–52. doi:10.1177/0013124508316309
Viteritti, J. P. 2010. "School Choice and Market Failure: How Politics Trumps Economics in Education and Elsewhere." *Journal of School Choice* 4(2): 203–21. doi:10.1080/15582159.2010.483923
Whitty, G., S. Power, and D. Halpin. 1998. *Devolution and Choice in Education*. Buckingham: Open University Press.
Wilson, W. J. 2012. *The Truly Disadvantaged: The Inner City, the Underclass, and Public Policy*, 2nd ed. Chicago and London: University of Chicago Press.
Wong, Y. C. 2013. "Education and Income Inequality in Hong Kong." *Hong Kong Economic Journal*. Retrieved from www.wangyujian.com/?p=3425&lang=en
World Education News and Reviews. 2010. "Education in Taiwan." Retrieved from http://wenr.wes.org/2010/05/wenr-may-2010-feature/
Wu, X. 2012. "School Choice with Chinese Characteristics." *Comparative Education* 48(3): 347–66. doi:10.1080/03050068.2012.656901
Xu, D., and X. Wu. 2015a. "School Allocation Policy and the Reverse Gender Gap in Academic Achievement: Evidence from a Quasi-Experiment in Hong Kong." PSC Research Report No. 15–851.
——. 2015b. "The Rise of the Second Generation: Aspirations, Motivations and Academic Success of Chinese Immigrants' Children in Hong Kong." PSC Research Report No. 15–852.
Yung, A. M. 2006. "The Policy of Direct Subsidy Scheme Schools in Hong Kong: Finance and Administration." *Hong Kong Teachers' Centre Journal* 5: 94–111. Retrieved from http://edb.org.hk/HKTC/download/journal/j5/P094_111.pdf
Zhang, Z. 2014. "Contingent Transition to Triumph: Socioeconomic Attainments of Chinese Immigrants' Children in Hong Kong." *Chinese Sociological Review* 46(4): 68–88. doi:10.2753/CSA2162-0555460404
Zhao, X., L. Zhang, and T. O. K. Sit. 2004. "Income Inequalities Under Economic Restructuring in Hong Kong." *Asian Survey* 44(3): 442–73. doi:10.1525/as.2004.44.3.442
Zhou, Y., Y. -L. Wong, and W. Li. 2015. "Educational Choice and Marketization in Hong Kong: The Case of Direct Subsidy Scheme Schools." *Asia Pacific Education Review* 16(4): 627–36. doi:10.1007/s12564-015-9402-9

About the Authors

Yisu Zhou is an assistant professor of educational policy at the University of Macau. His research focus is on teacher quality and teacher labor market. His

Downloaded by [University of Sussex Library] at 19:52 25 May 2016

expertise is in large-scale assessment and education development in China. His interests in education policy span across various topics, including educational finance, teacher education, economics of education, and social stratification and mobility. He is the co-author of the 2006 book *The Journey to Yellow Sheep River*, which discusses teachers' life in rural schools in China.

Tianji Cai is an assistant professor of sociology at the University of Macau. His research focuses on quantitative research methods, especially the issues of sampling weights in multilevel and longitudinal models. In addition, he is also interested in integrating genetics and sociology in the studies of social and health behaviors.

Dan Wang is an associate professor in the Faculty of Education at the University of Hong Kong. She has expertise in the sociology of education with a special interest in education in China. Her current research focuses on issues of educational inequality and teachers' work in China. Her recently publications include *The Demoralization of Teachers: Crisis in a Rural School in China* (Lexington Books) and journal articles in *Journal of Contemporary China, Teaching and Teacher Education*, and *International Journal of Chinese Education.*

Appendix

Table A1

Sample Description of Hong Kong

	Number of School		Number of students	
	Sample	Total	Sample	Total
	2006–2007			
Government	11	36	372	33,717
Aided	127	382	4,031	385,003
Private/international	1	57	24	20,289
DSS	7	55	218	41,766
	2009–2010			
Government	10	32	318	30,146
Aided	126	370	4,046	365,213
Private/international	2	60	61	22,984
DSS	13	61	412	51,123
	2012–2013			
Government	10	32	310	26,313
Aided	119	365	3,812	321,538
Private/international	3	61	44	21,951
DSS	16	61	504	48,985

Downloaded by [University of Sussex Library] at 19:52 25 May 2016